Canadian Reindeer PrioScout Report (Round 2)



Methodology

Swab pellets were re-suspended in 500µL of 0.1% SDS/1X PBS. Samples were run by RT-QuIC in 4 replicates for 72 hours at 42°C. Results were compared against the negative plate controls using a t-test with the alternative hypothesis being that a sample will have a larger value than the negative control. Metrics analyzed were maxpoint ratio (MPR), maximum slope (MS), and area under the curve (AUC). The p-values for each metric must have been ≤ 0.05 and at least 50% of replicates must have crossed a threshold of 3 times the background fluorescence for a sample to have been considered "detected". If a sample had at least 50% of its replicates cross the threshold, but was negative for any of the other three metrics, it was labelled as "suspect".

Swab Results

Positive sample IDs: **224**; Suspect sample IDs: **120, 147, 153, 160, 179**

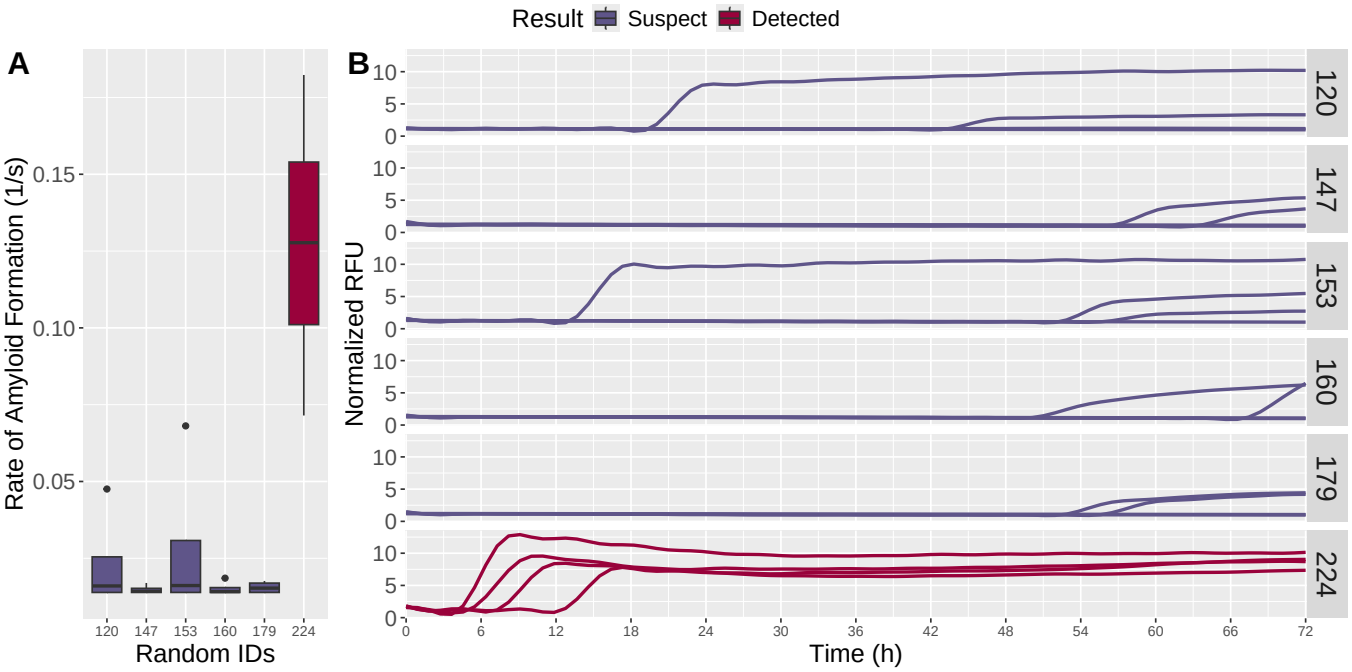


Figure 1: Positive or suspect samples. Samples are identified by the blinded 'Random ID' provided by the collection team. A: The rates of amyloid formation for the positive samples. B: The real-time graphs.

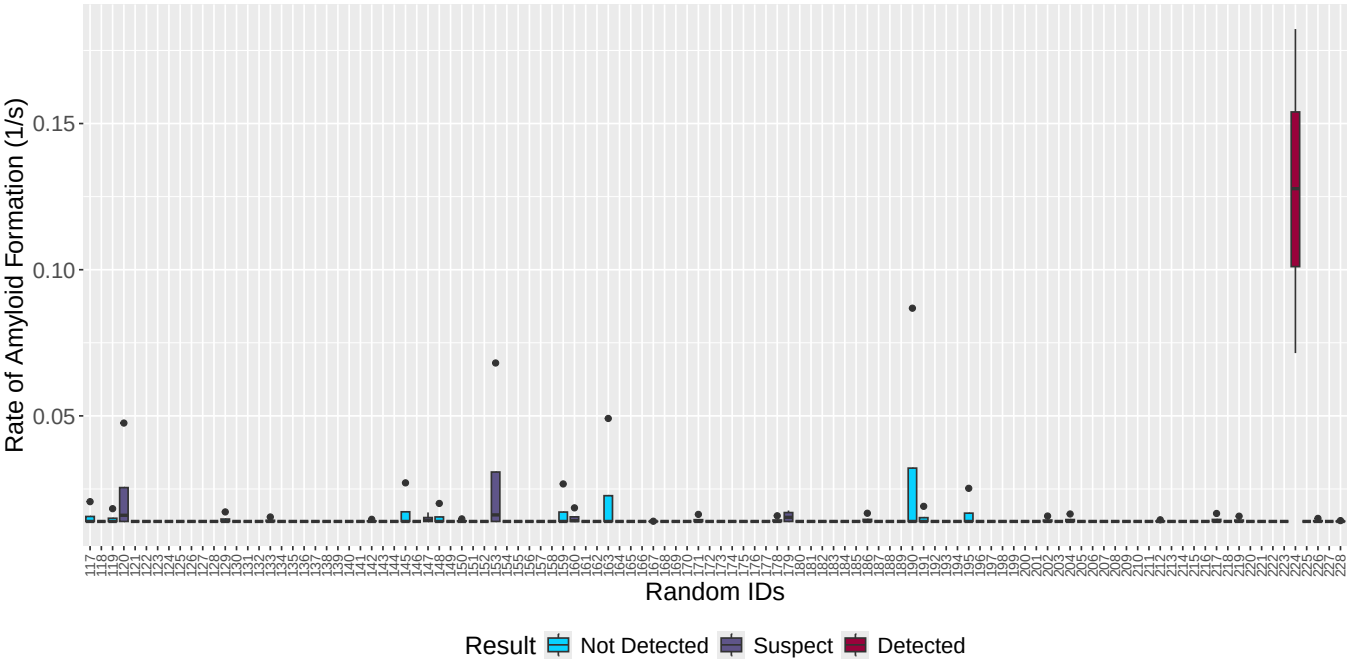


Figure 2: Rates of amyloid formation (RAF) of each sample. A larger RAF indicates a higher prion load in the sample.

Table 1: Results table of RT-QuIC testing. 'crossed': fraction of replicates above threshold; 'thres pos': at least half of replicates crossed threshold?; MPR: maxpoint ratio; MS: max slope; AUC: area under the curve.

Sample IDs	Random IDs	crossed	thres pos	MPR pvalue	MS pvalue	AUC pvalue	Result
S-2094	117	0.25	FALSE	0.202	0.178	0.205	Not Detected
S-2095	118	0	FALSE	0.59	0.528	0.636	Not Detected
S-2096	119	0.25	FALSE	0.365	0.319	0.311	Not Detected
S-2097	120	0.5	TRUE	0.142	0.128	0.141	Suspect
S-2098	121	0	FALSE	0.605	0.549	0.62	Not Detected
S-2099	122	0	FALSE	0.423	0.303	0.525	Not Detected
S-2100	123	0	FALSE	0.588	0.514	0.559	Not Detected
S-2101	124	0	FALSE	0.476	0.345	0.57	Not Detected
S-2102	125	0	FALSE	0.6	0.538	0.568	Not Detected
S-2103	126	0	FALSE	0.581	0.429	0.517	Not Detected
S-2104	127	0	FALSE	0.297	0.227	0.427	Not Detected
S-2105	128	0	FALSE	0.615	0.535	0.602	Not Detected
S-2106	129	0.25	FALSE	0.268	0.227	0.339	Not Detected
S-2107	130	0	FALSE	0.6	0.528	0.57	Not Detected
S-2108	131	0	FALSE	0.603	0.538	0.572	Not Detected
S-2109	132	0	FALSE	0.605	0.524	0.564	Not Detected
S-2110	133	0.25	FALSE	0.375	0.283	0.484	Not Detected
S-2111	134	0	FALSE	0.618	0.548	0.618	Not Detected
S-2112	135	0	FALSE	0.429	0.266	0.558	Not Detected
S-2113	136	0	FALSE	0.607	0.506	0.549	Not Detected
S-2114	137	0	FALSE	0.617	0.503	0.548	Not Detected
S-2115	138	0	FALSE	0.647	0.507	0.539	Not Detected
S-2116	139	0	FALSE	0.333	0.512	0.609	Not Detected
S-2117	140	0	FALSE	0.336	0.497	0.589	Not Detected
S-2118	141	0	FALSE	0.316	0.475	0.456	Not Detected
S-2119	142	0.25	FALSE	0.18	0.219	0.458	Not Detected
S-2120	143	0	FALSE	0.271	0.332	0.486	Not Detected
S-2121	144	0	FALSE	0.317	0.477	0.554	Not Detected
S-2122	145	0.25	FALSE	0.146	0.18	0.179	Not Detected
S-2123	146	0	FALSE	0.293	0.47	0.535	Not Detected
S-2124	147	0.5	TRUE	0.107	0.121	0.274	Suspect
S-2125	148	0.25	FALSE	0.12	0.132	0.157	Not Detected
S-2126	149	0	FALSE	0.327	0.468	0.484	Not Detected
S-2127	150	0.25	FALSE	0.231	0.313	0.376	Not Detected
S-2128	151	0	FALSE	0.266	0.264	0.423	Not Detected
S-2129	152	0	FALSE	0.364	0.494	0.541	Not Detected
S-2130	153	0.5	TRUE	0.0827	0.0894	0.129	Suspect
S-2131	154	0	FALSE	0.346	0.461	0.521	Not Detected
S-2132	155	0	FALSE	0.335	0.481	0.526	Not Detected
S-2133	156	0	FALSE	0.34	0.451	0.486	Not Detected

Table 1: Results table of RT-QuIC testing. 'crossed': fraction of replicates above threshold; 'thres pos': at least half of replicates crossed threshold?; MPR: maxpoint ratio; MS: max slope; AUC: area under the curve. *(continued)*

Sample IDs	Random IDs	crossed	thres pos	MPR pvalue	MS pvalue	AUC pvalue	Result
S-2134	157	0	FALSE	0.348	0.355	0.518	Not Detected
S-2135	158	0	FALSE	0.323	0.448	0.449	Not Detected
S-2136	159	0.25	FALSE	0.16	0.144	0.203	Not Detected
S-2137	160	0.5	TRUE	0.0875	0.112	0.194	Suspect
S-2138	161	0	FALSE	0.381	0.517	0.625	Not Detected
S-2139	162	0	FALSE	0.39	0.529	0.613	Not Detected
S-2140	163	0.25	FALSE	0.159	0.151	0.182	Not Detected
S-2141	164	0	FALSE	0.401	0.507	0.601	Not Detected
S-2142	165	0	FALSE	0.391	0.5	0.59	Not Detected
S-2143	166	0	FALSE	0.386	0.549	0.656	Not Detected
S-2144	167	0.25	FALSE	0.273	0.335	0.369	Not Detected
S-2145	168	0	FALSE	0.407	0.538	0.665	Not Detected
S-2146	169	0	FALSE	0.387	0.523	0.588	Not Detected
S-2147	170	0	FALSE	0.38	0.528	0.605	Not Detected
S-2148	171	0.25	FALSE	0.198	0.218	0.205	Not Detected
S-2149	172	0	FALSE	0.434	0.503	0.61	Not Detected
S-2150	173	0	FALSE	0.433	0.484	0.605	Not Detected
S-2151	174	0	FALSE	0.43	0.529	0.615	Not Detected
S-2152	175	0	FALSE	0.392	0.414	0.565	Not Detected
S-2153	176	0	FALSE	0.397	0.409	0.6	Not Detected
S-2154	177	0	FALSE	0.434	0.55	0.652	Not Detected
S-2155	178	0.25	FALSE	0.153	0.108	0.33	Not Detected
S-2156	179	0.5	TRUE	0.122	0.142	0.234	Suspect
S-2157	180	0	FALSE	0.325	0.321	0.571	Not Detected
S-2158	181	0	FALSE	0.407	0.502	0.552	Not Detected
S-2159	182	0	FALSE	0.44	0.521	0.578	Not Detected
S-2160	183	0	FALSE	0.421	0.552	0.636	Not Detected
S-2161	184	0	FALSE	0.284	0.245	0.494	Not Detected
S-2162	185	0	FALSE	0.339	0.407	0.49	Not Detected
S-2163	186	0.25	FALSE	0.285	0.356	0.333	Not Detected
S-2164	187	0	FALSE	0.232	0.262	0.293	Not Detected
S-2165	188	0	FALSE	0.314	0.185	0.477	Not Detected
S-2166	189	0	FALSE	0.416	0.548	0.627	Not Detected
S-2167	190	0.25	FALSE	0.196	0.228	0.207	Not Detected
S-2168	191	0.25	FALSE	0.209	0.218	0.256	Not Detected
S-2169	192	0	FALSE	0.313	0.375	0.436	Not Detected
S-2170	193	0	FALSE	0.314	0.291	0.494	Not Detected
S-2171	194	0	FALSE	0.489	0.568	0.636	Not Detected
S-2172	195	0.25	FALSE	0.202	0.236	0.231	Not Detected
S-2173	196	0	FALSE	0.47	0.56	0.619	Not Detected
S-2174	197	0	FALSE	0.482	0.566	0.659	Not Detected
S-2175	198	0	FALSE	0.384	0.331	0.623	Not Detected

Table 1: Results table of RT-QuIC testing. 'crossed': fraction of replicates above threshold; 'thres pos': at least half of replicates crossed threshold?; MPR: maxpoint ratio; MS: max slope; AUC: area under the curve. *(continued)*

Sample IDs	Random IDs	crossed	thres pos	MPR pvalue	MS pvalue	AUC pvalue	Result
S-2176	199	0	FALSE	0.403	0.409	0.576	Not Detected
S-2177	200	0	FALSE	0.461	0.537	0.642	Not Detected
S-2178	201	0	FALSE	0.426	0.284	0.646	Not Detected
S-2179	202	0.25	FALSE	0.202	0.158	0.197	Not Detected
S-2180	203	0	FALSE	0.425	0.406	0.587	Not Detected
S-2181	204	0.25	FALSE	0.31	0.345	0.303	Not Detected
S-2182	205	0	FALSE	0.253	0.491	0.52	Not Detected
S-2183	206	0	FALSE	0.249	0.508	0.575	Not Detected
S-2184	207	0	FALSE	0.237	0.488	0.436	Not Detected
S-2185	208	0	FALSE	0.253	0.501	0.465	Not Detected
S-2186	209	0	FALSE	0.205	0.287	0.405	Not Detected
S-2187	210	0	FALSE	0.245	0.498	0.493	Not Detected
S-2188	211	0	FALSE	0.245	0.46	0.446	Not Detected
S-2189	212	0.25	FALSE	0.168	0.251	0.402	Not Detected
S-2190	213	0	FALSE	0.248	0.515	0.513	Not Detected
S-2191	214	0	FALSE	0.24	0.489	0.484	Not Detected
S-2192	215	0	FALSE	0.275	0.476	0.436	Not Detected
S-2193	216	0	FALSE	0.302	0.496	0.5	Not Detected
S-2194	217	0.25	FALSE	0.195	0.289	0.331	Not Detected
S-2195	218	0	FALSE	0.286	0.503	0.539	Not Detected
S-2196	219	0.25	FALSE	0.199	0.297	0.349	Not Detected
S-2197	220	0	FALSE	0.289	0.472	0.498	Not Detected
S-2198	221	0	FALSE	0.283	0.511	0.548	Not Detected
S-2199	222	0	FALSE	0.28	0.49	0.492	Not Detected
S-2200	223	0	FALSE	0.28	0.494	0.521	Not Detected
S-2201	224	1	TRUE	0.027	0.0262	0.0254	Detected
S-2202	225	0	FALSE	0.296	0.497	0.467	Not Detected
S-2203	226	0.25	FALSE	0.159	0.199	0.354	Not Detected
S-2204	227	0	FALSE	0.253	0.514	0.511	Not Detected
S-2205	228	0.25	FALSE	0.157	0.167	0.293	Not Detected